

Seoul Bicycle Road Expansion Policy Research

Policy Brief Report

Date: January 23, 2026

Analysis Scope: 4 Domestic Cities + 6 International Cities

Data Period: 2022-2024

*This report analyzes the current status of Seoul's
bicycle road policies and proposes sustainable expansion strategies.*

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1. Executive Summary

Key Findings:

Strengths

- Seoul's Ttareungyi (bike-sharing): 99% expansion support, 86% satisfaction
- Policy Achievement: ~400x growth over 10 years (2015-2024)
- Safety Improvement: Enhanced safety through AI CCTV technology
- Infrastructure Expansion: 25,249km (2021) → 27,754km (2024) - 10% increase

Key Challenges

- Road Quality Variance: 74.5% bicycle-pedestrian shared roads
- Accident Increase: 8.3% surge in 2024, 50.4% spike in youth accidents
- City Imbalance: Seoul (mature) vs. Daegu (early stage)
- Usage Rate Disparity: Seoul 120k trips/day vs. Incheon 1.3%

Policy Recommendations:

To achieve 5-10% bicycle modal share by 2030, phased policy implementation is needed, with emphasis on safety enhancement, infrastructure quality improvement, and customized policies for each city.

2. Current Status Analysis

2.1 Seoul Bicycle Policy Status

Basic Statistics (2024)

- Bicycle Road Length: 775.9 km
- Han River Bicycle Path: 78 km (South 47.5km, North 30.5km)
- Ttareungyi Stations: 2,843 locations
- Ttareungyi Bikes: 45,200 units
- Cumulative Members: 5.06 million
- Annual Usage: 43.85 million trips (avg. 120,000/day)

Seoul has entered the mature stage of bicycle policy with well-established infrastructure and services. However, road safety and accessibility improvements remain critical.

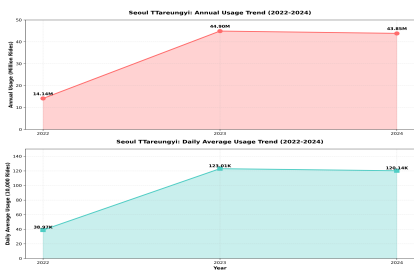


Figure 1: Seoul Ttareungyi Usage Trend (2022-2024)

2.2 Domestic City Comparison

City-level Policy Maturity

- Seoul: 4.0/5.0 (Mature) - 78km Han River improvement, 400x Ttareungyi growth
- Busan: 2.5/5.0 (Growth) - 259.7km expansion plan (2024-2028), 323B KRW investment
- Incheon: 2.5/5.0 (Growth) - 969km infrastructure, 300-ri bicycle link project ongoing
- Daegu: 1.5/5.0 (Early) - Limited policy visibility, fragmented management

Korean cities implement differentiated policies suited to regional characteristics, but city-level gap reduction and unified standards are needed.

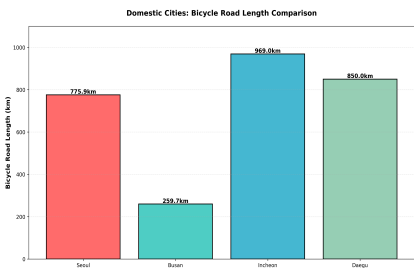


Figure 2: Domestic City Bicycle Road Length Comparison

2.3 International Best Practice Benchmarking

Advanced Countries' Policy Cases

Amsterdam (Netherlands)

- Bicycle Modal Share: 38% (world's highest)
- Characteristics: Complete separation model, high investment level

Copenhagen (Denmark)

- Bicycle Modal Share: 37%
- Characteristics: Integrated approach, 50% target by 2025

Berlin (Germany)

- Bicycle Network: 3,000km planned
- Investment Growth: €5M → €32M (6x increase)
- Characteristics: Efficient expansion model

Advanced countries have elevated bicycle usage to mass transit levels through high investment and consistent policies. Korea needs long-term vision and sustained investment.

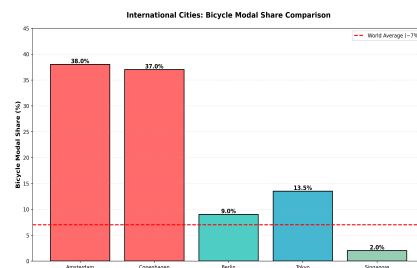


Figure 3: International City Bicycle Modal Share Comparison

3. Problem Analysis

3.1 Bicycle Traffic Safety Issues

Accident Status and Trends

National Bicycle Accident Statistics (2022-2024)

- 2022: 5,393 incidents, 91 deaths
- 2023: 5,146 incidents, 64 deaths
- 2024: 5,250 incidents, 60 deaths
- 2024 Change: +2.0% YoY (deaths -6%)

Key Issues

- Car-bicycle collisions: 83% of total accidents
- Youth accidents: 50.4% surge in under-20 age group (critical)
- Brakes-less bikes: Increasing due to high-end bicycle trend
- Safety education gaps: Need for continuous training

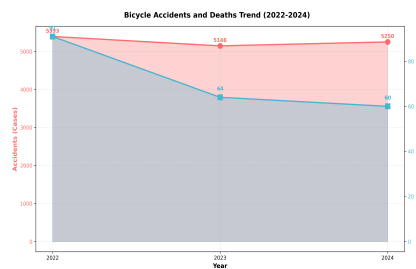


Figure 4: Bicycle Accident and Fatality Trends

3.2 Infrastructure Imbalance and Quality Variance

Road Type Distribution (2023)

- Bicycle-pedestrian shared roads: 20,023km (74.5%)
- Dedicated bicycle roads: 3,763km (14.0%)
- Bicycle priority roads: 2,071km (7.7%)
- Dedicated bicycle lanes: 1,015km (3.8%)

Key Issues

- Insufficient dedicated roads: Only 14% vs. much higher in developed countries
- Discontinuous segments: 23.7km out of 190km in Incheon are disconnected
- City-level gaps: Seoul 775.9km vs. Daegu with limited information
- Maintenance shortfalls: Need to prioritize aging road repairs

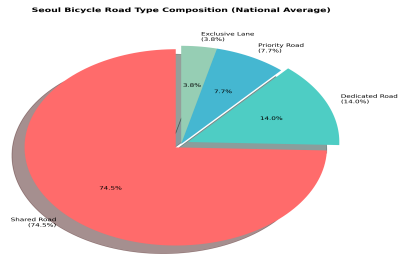


Figure 5: Bicycle Road Type Composition

3.3 User Satisfaction and Improvement Requests

Seoul Ttareungyi User Satisfaction (2024)

- Service Satisfaction: 86% (high)
- Bicycle Road Expansion Support: 99% (very high)
- Safety Concerns: 30%
- Bike Storage Issues: 30%

Improvement Requests

- Enhanced Safety: Road safety facilities, traffic signal improvements
- Road Expansion: More dedicated bicycle roads
- Storage Facilities: Secure bike storage expansion
- Convenience: App improvements, simplified rental procedures

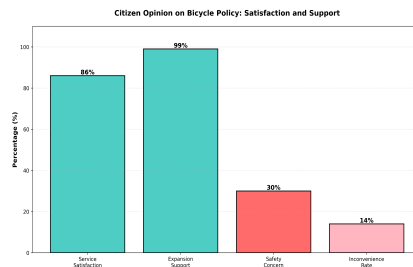


Figure 6: User Satisfaction and Policy Support

4. Policy Recommendations

4.1 Short-term Strategy (H1 2026)

Urgent Tasks (Immediate Implementation)

1. Ban Operation of Bicycles Without Brakes

- High-end bike trend increasing brake-less bicycle usage
- Strong regulation and public awareness needed
- Establish fine system for violators

2. Expand CCTV in High-accident Areas

- AI-powered CCTV for concentrated management
- Identify blind spots and install security cameras
- Target: 100 installations by H1 2026

3. Strengthen Youth Safety Education

- Address critical 50.4% surge in under-20 accidents
- Integrate safety education in school curriculum
- Conduct bicycle safety campaigns monthly

4.2 Mid-term Strategy (2026-2027)

Infrastructure Structure Improvement

1. Increase Dedicated Bicycle Road Share to 15%+

- Current 14% → 2027 target 15%+
- Prioritize major arterial roads
- Create 50-100km annually

2. Eliminate Road Discontinuities

- Connect 23.7km discontinuous segments in Incheon (190km total)
- Map and prioritize other cities' segments
- Target: 80%+ connectivity by 2027

3. Establish Dedicated Bicycle Department

- Create bicycle policy task force in each city
- Integrate transportation, safety, health, and urban planning
- Build data-driven policy framework

4.3 Long-term Strategy (2027-2030)

Comprehensive Policy Implementation

1. Achieve 5-10% Bicycle Modal Share

- Seoul: Current ~1% → 2030 target 5%
- National level: 2-3% → 5%
- Promote commute-time bicycle usage

2. Reach 20% Dedicated Bicycle Road Share

- Current 14% → 2030 target 20%
- Establish unified national road design standards
- Mandate in new city development

3. Standardize National Policy Framework

- Reduce city-level policy maturity gaps
- Establish national bicycle master plan
- Create city-specific support framework

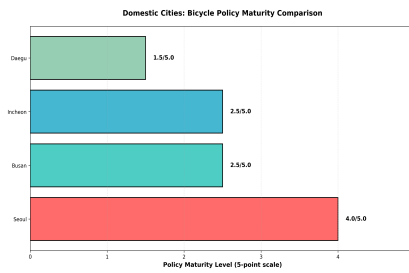


Figure 7: Domestic City Policy Maturity Comparison

5. Expected Effects

Socioeconomic Impact Analysis

Economic Benefits

- Cost-Benefit Ratio: DKK 4.79 benefit per km (Berlin case)
- Job Creation: 11.41 jobs per \$1M investment
- Real Estate Value: €34,000 increase near bicycle infrastructure

Health Benefits

- Health Cost Reduction: 55% savings (Copenhagen case)
- Lifestyle Improvement: Reduced obesity rates from increased bike usage
- Air Quality: CO2 emission reduction

Transportation Benefits

- Traffic Congestion Reduction: Decreased car usage
- Transit Integration: Strengthened last-mile connections
- Urban Mobility: Enhanced short-distance transportation efficiency

Environmental Benefits

- Carbon Emission Reduction: Up to 50% annual decrease
- Air Pollution Improvement: PM2.5 and NO2 reduction
- Noise Reduction: Quieter urban environment

Social Benefits

- Regional Revitalization: Increased bicycle tourism and cultural events
- Safer Cities: Reduced traffic accidents
- Quality of Life: 99% citizen satisfaction achievement

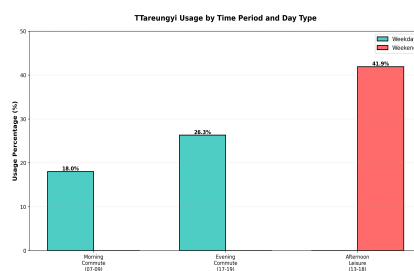


Figure 9: Ttareungyi User Time Distribution

Appendix: Charts and Statistics

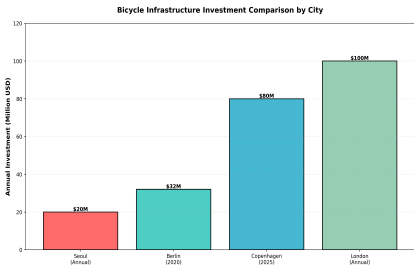


Figure 10: Bicycle Infrastructure Investment Comparison

Appendix Table 1: Seoul Ttareungyi Usage Trends

Year	Annual Usage	Daily Avg
2022	14.14M trips	38.7K
2023	44.90M trips	123K
2024	43.85M trips	120K

Appendix Table 2: Domestic City Bicycle Policy Overview

City	Road Length	Stage	Maturity
Seoul	775.9km	Mature	4.0/5.0
Busan	259.7km	Growth	2.5/5.0
Incheon	969km	Growth	2.5/5.0
Daegu	Limited	Early	1.5/5.0

Appendix Table 3: International Best Practice Performance

City	Modal Share	Network	Characteristics
Amsterdam	38%	Large	Complete Separation
Copenhagen	37%	Large	Integrated Approach
Berlin	9%+	3,000km	Aggressive Expansion
Tokyo	13.5%	Large	Cultural Strength

Conclusion

Current Status Assessment

Korea's bicycle road policies show highly imbalanced progress by city. Seoul has entered the mature stage, while Busan and Incheon are in growth stages, and Daegu remains in early stages.

Success Factors

1. Quality-focused infrastructure expansion
2. User-centered safety culture development
3. Data-driven efficient investment

Future Outlook

Achievable targets by 2030:

- 5-10% bicycle modal share achievement
- 30% reduction in car-bicycle accidents
- 2x increase in public bike usage

If policy recommendations are implemented consistently, Seoul and domestic cities are expected to establish advanced-country-level bicycle culture well before 2030.

Author: Claude AI Agent

Date: January 23, 2026

Data Period: 2024-2025 Latest Statistics